Infectious Disease Simulation

Part I: Explore the VaxImpact Simulator here: https://herdimmunitysimulation.base44.app/
Spend a few minutes freely exploring the simulator and writing down notes about the following

1. What variables can you manipulate in this simulator?

2. Describe one change (other than vaccine rates) you made that significantly altered the outbreak curve. What happened?

Part II: Measles Case Study (Vaccination Rate Experiment)
Measles is a highly contagious virus and it takes between 10 to 14 days to recover. Although the mortality rate is low (approximately 1%), measles is considered a very serious disease because of how easily it can spread between individuals. The vaccine for measles is very effective (97%) after two doses.

In the following experiment, you will test to see how vaccination rates influence the spread of measles in a community (such as a school/classroom).

Set the simulator to the following parameters:
Transmission Rate: 90%
Recovery time: 14 days (2 weeks)
Mortality Rate: 1%
Population Density: 80%
Vaccine Effectiveness: 97%
Vaccination Rates: This will be the independent variable (changed between experiments)
NOTE: The simulation ends automatically. Just hit play and wait for the simulation to play out

Table 1: Measles Cases with Vaccination Rates set to 65%

Trial	Maximum Sick People in a single day (Peak Infections)	Total infected	% Vaccinated sick	% Unvaccinated sick
1				
2				
3				
Average				

Table 2: Measles Cases with Vaccination Rates set to 80%

Trial	Maximum Sick People in a single day (Peak Infections)	Total infected	% Vaccinated sick	% Unvaccinated sick
1				
2				
3				
Average				

Table 3: Measles Cases with Vaccination Rates set to 95%

Trial	Maximum Sick People in a single day (Peak Infections)	Total infected	% Vaccinated sick	% Unvaccinated sick
1				
2				
3				
Average				

Analysis Questions:

Note: In the top right corner of the simulator, there is a learn more button to provide you with more details about each variable. You may use this to help you with some of the following questions

1. How did increasing the vaccination rate affect outbreak size? Be sure to include some of the results from your experiment.

2. How may vaccine effectiveness influence your results? How might you test this with the simulator?

3. Many diseases can spread through indirect contact with others, such as germs living on door handles for days. How might you predict this to change the results of the simulator?

4. Based on your data, explain the concept of **herd immunity** in your own words.

5. One important variable not included in this simulation is hospital burden. How might vaccination rates indirectly influence hospital burden (number of sick and hospitalized people at any given time)? Would you expect this to also affect the mortality rate if hospitals and healthcare professionals are unable to manage the volume of cases occurring within a single day or season?

6. This simulator assumes that people in a community (such as a school) are isolated from everyone else and can only become infected by others within that community. Would you expect such an isolated scenario to occur in real life? Explain why or why not, and how this might alter patterns of disease transmission.